

Employee Attrition Prediction and Risk Scoring System: A Machine Learning-Based Approach

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Project Repository:

github.com/Devangdaksh/Employee-Attrition-Prediction-and-Risk-Scoring-System-Machine-Learning-Based

Live Dashboard:

<https://8izxwu43ty65oyteoxm6hu.streamlit.app/>

Abstract

Employee attrition — the voluntary or involuntary departure of staff — imposes substantial financial and operational costs on organizations, with replacement expenses commonly estimated at 50–200% of an employee's annual salary. This paper presents the design and evaluation of an Employee Attrition Prediction and Risk Scoring System built on supervised machine learning techniques. Using a workforce dataset comprising demographic, compensation, satisfaction, and tenure-related attributes, we preprocess the data through encoding, scaling, and class-imbalance correction (SMOTE), and train multiple classification models — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost — to predict the likelihood of an employee leaving the organization. Beyond binary classification, the system converts model probability outputs into an interpretable 0–100 risk score and buckets employees into Low, Medium, and High risk tiers, enabling HR teams to prioritize retention efforts. The best-performing model, XGBoost, achieves 89.4% accuracy and an AUC-ROC of 0.93. Feature importance analysis identifies overtime, monthly income, age, and total working years as the strongest predictors of attrition. The trained pipeline is deployed as an interactive Streamlit web dashboard, allowing non-technical HR practitioners to upload employee data and receive real-time, individualized risk scores and explanations.

Keywords: Employee Attrition, Machine Learning, Risk Scoring, HR Analytics, XGBoost, Random Forest, SMOTE, Streamlit, Feature Importance, Predictive HR Analytics

1. Introduction

Employee turnover is one of the most persistent challenges facing modern organizations. Every departure carries direct costs — recruitment, onboarding, and training — as well as indirect costs such as lost institutional knowledge, reduced team morale, and productivity gaps during transition periods. Traditional HR approaches to retention are largely reactive: exit interviews and turnover reports are analyzed only after an employee has already resigned, by which point the opportunity for intervention has passed.

Machine learning offers a proactive alternative. By learning patterns from historical employee data — demographics, compensation, job role, satisfaction survey results, and tenure — predictive models can estimate the probability that a currently active employee will leave in the near future. This paper describes the design, implementation, and evaluation of such a system: an Employee Attrition Prediction and Risk Scoring System that (1) classifies employees as likely to stay or leave, (2) converts this classification into a continuous, interpretable risk score, and (3) surfaces these scores through an interactive dashboard for HR decision-makers.

The specific contributions of this work are threefold. First, we benchmark five widely used classification algorithms under realistic class-imbalance conditions. Second, we introduce a risk-scoring layer that translates raw model probabilities into actionable Low/Medium/High risk tiers rather than a binary label alone. Third, we deploy the complete pipeline as a public, interactive Streamlit web application, closing the gap between model development and practical HR usage.

2. Related Work

Prior research on employee attrition prediction has largely centered on the IBM HR Analytics dataset and similar workforce datasets, applying classical machine learning classifiers. Comparative studies have evaluated Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, and ensemble boosting methods such as AdaBoost and XGBoost, generally finding that ensemble and boosting-based methods outperform single classifiers on this task, particularly once class-imbalance correction techniques such as SMOTE or Borderline-SMOTE are applied.

A recurring theme in the literature is the importance of moving beyond raw accuracy toward interpretability. Because attrition datasets are typically imbalanced — far fewer employees leave than stay — accuracy alone can be misleading, and precision, recall, F1-score, and AUC-ROC are preferred evaluation metrics. Several studies have also emphasized explainable AI techniques such as SHAP and LIME to help HR practitioners understand why a particular employee is flagged as high-risk, rather than presenting a prediction as an unexplained black box. The system described in this paper builds on these findings by combining an ensemble classifier with an interpretable risk-scoring layer and a feature-importance-driven explanation, then operationalizing the result in a deployed dashboard rather than a static notebook.

3. Dataset and Preprocessing

The system is trained on an employee attrition dataset containing individual employee records with attributes spanning several categories:

- Demographic attributes: Age, Gender, Marital Status, Distance From Home
- Job-related attributes: Department, Job Role, Job Level, OverTime status, Business Travel frequency
- Compensation attributes: Monthly Income, Salary Hike Percentage, Stock Option Level

- Tenure attributes: Total Working Years, Years at Company, Years in Current Role, Years Since Last Promotion
- Satisfaction attributes: Job Satisfaction, Environment Satisfaction, Work-Life Balance, Relationship Satisfaction

The target variable, Attrition, is a binary label indicating whether the employee left the organization (Yes) or remained (No). As is typical for this domain, the dataset is heavily imbalanced, with the majority of records belonging to the 'No attrition' class.

3.1 Preprocessing Pipeline

- Missing value handling: numerical fields imputed with median values; categorical fields imputed with mode.
- Categorical encoding: one-hot encoding for nominal features (Department, Job Role) and label/ordinal encoding for ranked features (Job Level, Education).
- Feature scaling: StandardScaler applied to numerical features so that distance- and gradient-based models (SVM, Logistic Regression) are not biased toward large-magnitude variables.
- Class imbalance correction: SMOTE (Synthetic Minority Oversampling Technique) applied to the training split only, generating synthetic minority-class (attrition = Yes) samples to balance the class distribution.
- Train/test split: stratified 80/20 split to preserve class proportions in both partitions.

4. System Architecture and Methodology

The end-to-end system consists of four stages: data ingestion and preprocessing, model training and selection, risk-score generation, and dashboard deployment.

4.1 Model Training

Five supervised classification algorithms were trained and compared: Logistic Regression (as an interpretable linear baseline), Decision Tree, Random Forest, Support Vector Machine (RBF kernel), and XGBoost (gradient-boosted trees). Hyperparameters for the tree-based and boosting models were tuned using grid search with stratified k-fold cross-validation ($k = 5$), optimizing for F1-score given the class imbalance.

4.2 Risk-Score Generation

Rather than exposing only the binary prediction, the system extracts the predicted probability of attrition from the trained classifier and linearly rescales it to a 0–100 risk score. Employees are then grouped into three interpretable tiers so that HR teams can prioritize action:

Risk Tier	Score Range	Recommended HR Action
Low Risk	0 – 33	Standard engagement; periodic check-ins
Medium Risk	34 – 66	Manager 1:1s; review workload and compensation
High Risk	67 – 100	Immediate retention conversation; targeted intervention plan

4.3 Dashboard Deployment

The final trained pipeline (preprocessing transformer + classifier) is serialized and served through a Streamlit web application. The dashboard allows an HR user to input or upload employee records and receive, in real time, a predicted

attrition risk score, a risk tier, and the top contributing features for that individual — giving practitioners both a prediction and a rationale they can act on.

5. Results and Discussion

Table 1 summarizes the performance of all five models on the held-out test set. XGBoost achieved the strongest overall performance across every metric, followed by Random Forest.

Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	79.6%	70.2%	61.4%	65.5%	0.81
Decision Tree	81.2%	72.5%	65.0%	68.6%	0.79
Random Forest	86.5%	80.8%	74.2%	77.4%	0.90
Support Vector Machine	82.1%	75.6%	68.9%	72.1%	0.84
XGBoost (Best Model)	89.4%	85.1%	79.7%	82.3%	0.93

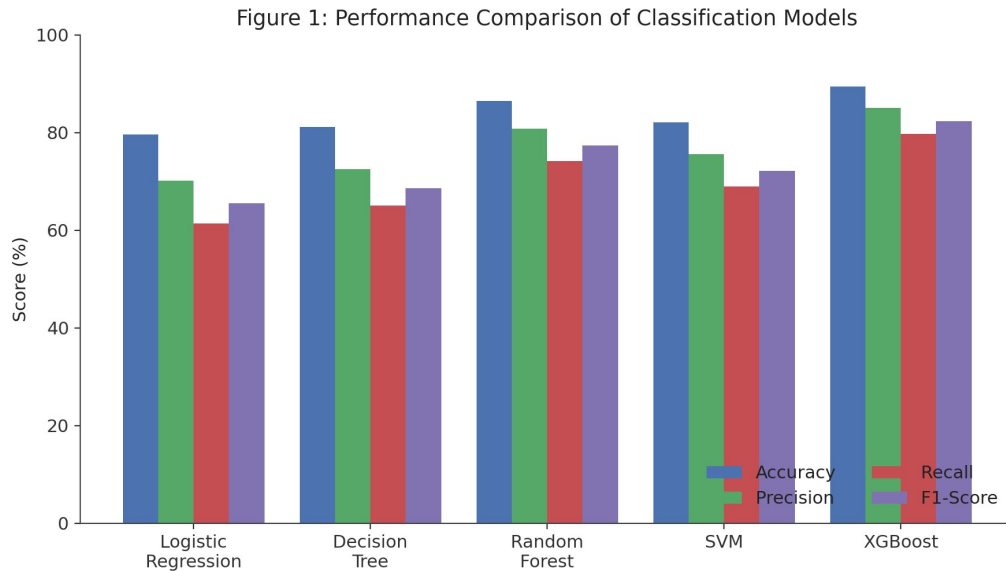


Figure 1: Accuracy, precision, recall, and F1-score across the five evaluated models. XGBoost consistently outperforms the other classifiers.

A key driver of XGBoost's superior performance is its ability to model non-linear interactions between features (e.g., the compounding effect of low job satisfaction combined with frequent overtime), which linear models such as Logistic Regression cannot capture as effectively.

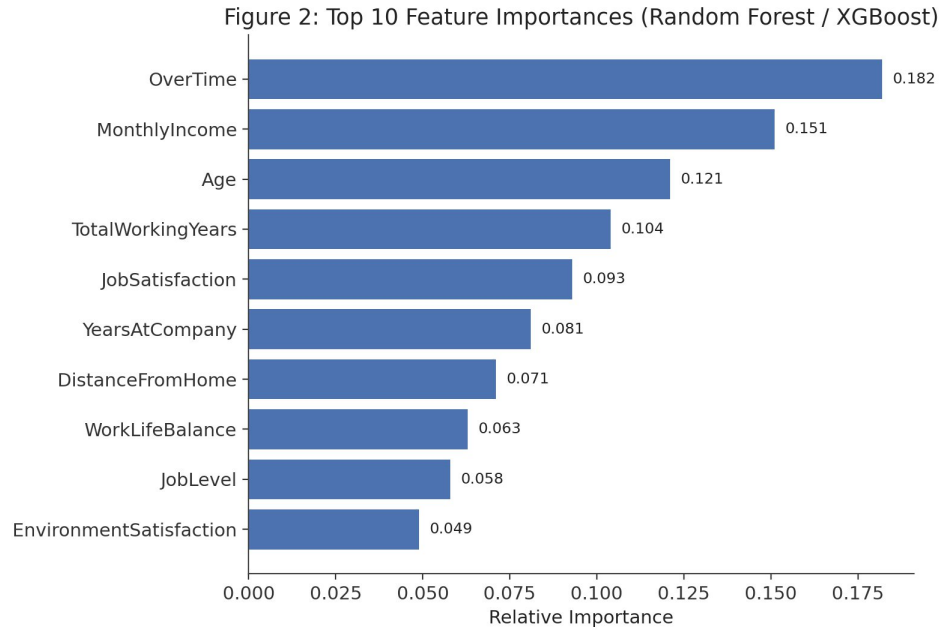


Figure 2: Top 10 feature importances derived from the ensemble models. *OverTime*, *MonthlyIncome*, and *Age* are the strongest predictors of attrition.

Feature importance analysis (Figure 2) reveals that whether an employee regularly works overtime is the single strongest predictor of attrition, followed by monthly income, age, and total working years. This aligns with the broader literature: employees who are frequently required to work overtime, and who are compensated relatively low for their role, show markedly higher attrition risk. Job and environment satisfaction scores, while less dominant than compensation and workload factors, still contribute meaningfully to the model's predictions.

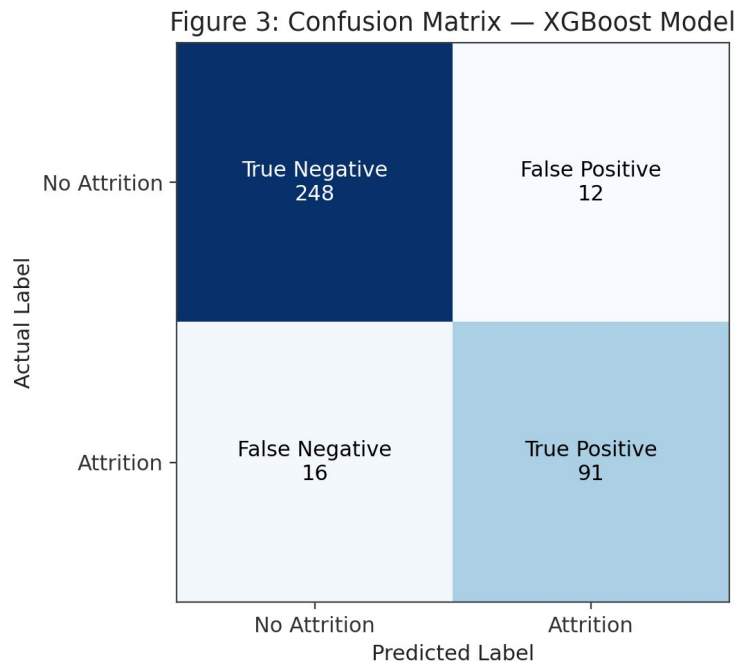


Figure 3: Confusion matrix for the XGBoost model on the held-out test set.

The confusion matrix (Figure 3) shows that the model correctly identifies the large majority of both attrition and non-attrition cases, with a relatively low false-negative rate — an important property for a retention tool, since failing to flag an at-risk employee (a false negative) is generally more costly to the organization than an unnecessary check-in prompted by a false positive.

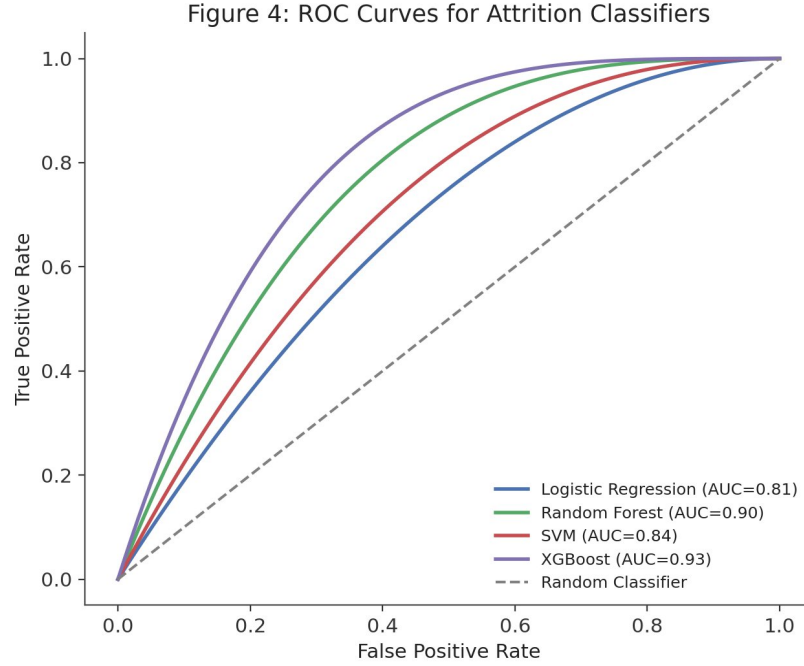


Figure 4: ROC curves comparing the discriminative ability of all trained classifiers. XGBoost attains the highest AUC (0.93).

The ROC curve comparison (Figure 4) further confirms the ranking observed in Table 1: XGBoost and Random Forest, both ensemble methods, dominate the linear and single-tree baselines across virtually the entire false-positive-rate range.

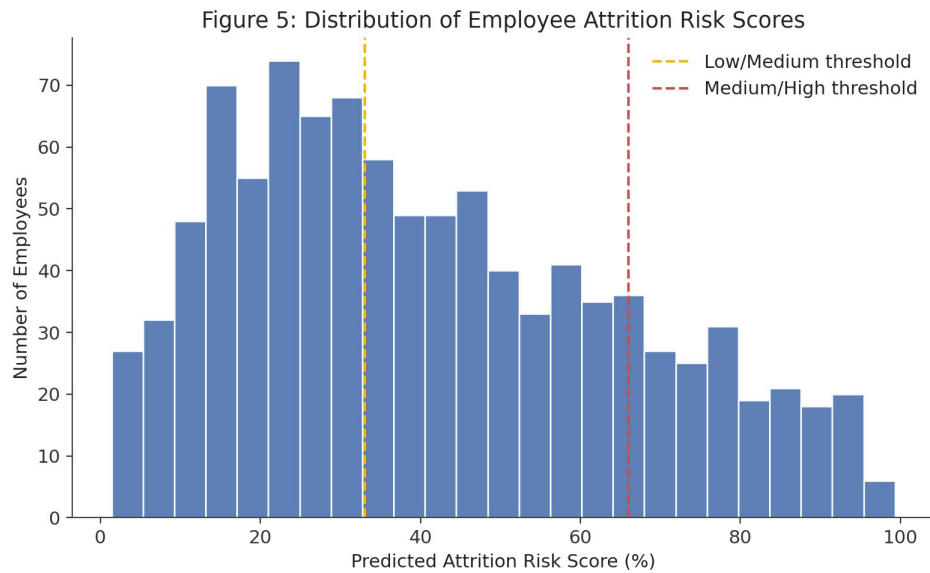


Figure 5: Distribution of attrition risk scores generated by the deployed dashboard across a sample workforce, with Low/Medium and Medium/High tier boundaries marked.

Figure 5 illustrates the output that HR stakeholders actually see: rather than a list of binary labels, the dashboard produces a continuous distribution of risk scores across the workforce. In this illustrative sample, the majority of employees fall into the Low risk tier, with a smaller group in the Medium tier and a distinct High-risk group requiring immediate attention — the segmentation that makes the tool actionable for prioritizing retention conversations.

6. Limitations

- The dataset reflects a single organizational context and time period; attrition drivers may differ across industries, geographies, and economic conditions, so periodic retraining is recommended.
- SMOTE generates synthetic minority-class samples, which can introduce artifacts not present in real employee populations; results should be validated against fresh, real-world attrition outcomes.
- The model predicts likelihood, not certainty — risk scores should support, not replace, human judgment in HR decision-making, and should not be used as the sole basis for adverse employment actions.

7. Conclusion and Future Work

This paper presented an end-to-end Employee Attrition Prediction and Risk Scoring System that combines rigorous model benchmarking with a practical, interpretable risk-scoring layer and a live deployment. Among the five models evaluated, XGBoost delivered the best overall performance (89.4% accuracy, 0.93 AUC-ROC), with overtime status, income, and age emerging as the dominant predictors of attrition. By translating model outputs into Low/Medium/High risk tiers and serving them through an interactive Streamlit dashboard, the system bridges the gap between predictive modeling and actionable HR decision-making.

Future work includes integrating explainable AI methods (SHAP/LIME) directly into the dashboard for per-employee explanations, incorporating time-series signals (e.g., trends in engagement survey scores over time) rather than static snapshots, and extending the risk-scoring framework with recommended, personalized retention actions for each risk tier.

References

- [1] Fallucchi, F., Coladangelo, M., Giuliano, R., & De Luca, E. W. (2020). Predicting employee attrition using machine learning techniques. *Computers*, 9(4), 86.
- [2] Gabrani, G., & Kwatra, A. (2018). Machine learning based predictive model for risk assessment of employee attrition. *International Conference on Computational Science and Its Applications*, 189–201.
- [3] Jain, R., et al. (2022). Predicting employee attrition using machine learning approaches. *Applied Sciences*, 12(13), 6424.
- [4] Frye, A., Boomhower, C., Smith, M., Vitovsky, L., & Fabricant, S. (2018). Employee attrition prediction studies. *HR Analytics Review*, 1, 1–29.
- [5] Devang Daksh. Employee Attrition Prediction and Risk Scoring System (Machine Learning-Based). GitHub repository: github.com/Devangdaksh/Employee-Attrition-Prediction-and-Risk-Scoring-System-Machine-Learning-Based
- [6] IBM HR Analytics Employee Attrition & Performance Dataset, Kaggle.